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AI in education, sustainability, and the future of work: An integrative review of industry 5.0, education 5.0, and work 5.0

Bianca Ifeoma Chigbu *, Sicelo Leonard Makapela

Faculty of Law, Humanities and Social Sciences, Walter Sisulu University, South Africa

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ABSTRACT

Industry 5.0, Education 5.0, and Work 5.0 are often discussed in silos; this integrative review shows how they cohere into a human-centric, sustainability-aligned model of AI-enabled transformation. This study reviews 200 records (62 included) from Scopus, Web of Science, Google Scholar, and policy portals (EC, UNESCO, ILO, WEF) to examine how these three domains intersect as a unified paradigm. Using an integrative literature review with quality appraisal of empirical studies and policy reports, we identify three cross-domain themes: (1) human-AI collaboration that reorients automation from substitution to augmentation, (2) skills development pathways led by Education 5.0 that bridge education, industry, and labor, and (3) sustainability as a unifying normative anchor across ecological, social, and economic dimensions. Findings show benefits (ergonomics, curriculum-industry alignment, green jobs) and risks (deskilling, inequality, rebound effects). The framework contributes by clarifying shared mediators (skills, ethics, sustainability), surfacing contradictions (automation vs. inclusion), and proposing governance conditions for human-centric outcomes. Major implications include aligning innovation policy, curriculum reform, and labour protections under SDG-coherent strategies, and piloting metrics that capture well-being, equity, and productivity.

1. Introduction

In recent years, a constellation of “5.0” paradigms has emerged across industry, education, and work, each pointing to a new stage of evolution beyond the technology-centric focus of the “4.0” era. Industry 4.0, characterized by automation, artificial intelligence (AI), and cyber-physical systems, brought unprecedented digital transformation to manufacturing and services. However, as the fourth industrial revolution matured (Caiado et al., 2024; Piccarozzi et al., 2024), calls grew for a more human-centric and sustainable approach to innovation (European Commission, 2021; Grego et al., 2024; Grybauskas and Cárdenas-Rubio, 2024; Shahidi Hamedani et al., 2024). While Industry 5.0, Education 5.0, and Work 5.0 have each been introduced in isolation, what remains underdeveloped is a framework that explains how they intersect, reinforce, or contradict one another. The originality of this article lies in providing that synthesis: we move beyond descriptive accounts to critically interrogate how these paradigms overlap, where tensions emerge (e.g., between efficiency and inclusion, or innovation and sustainability), and what a unified “5.0” framework implies for future theory and practice.

In response, Industry 5.0 has been introduced as a vision of industrial development (Adel, 2022; Rejeb et al., 2025) that “aims beyond efficiency and productivity as the sole goals” and instead prioritizes societal value, placing worker well-being at the center of production while respecting planetary boundaries (European Commission, 2021; Gola et al., 2025). This concept, first articulated by the European Commission in 2021, represents a shift to human-centric innovation using advanced technologies not to replace humans but to empower workers and solve global challenges. In this review, automation denotes programmable mechanization and algorithmic control spanning the 4.0 and 5.0 eras. What differentiates 5.0 is not the presence of automation but its purpose and governance: 5.0 embeds automation within a human-centric, augmentation-first design (human-in-the-loop by default, participation, safety, and sustainability metrics), whereas 4.0 is often framed as automation-first for productivity. We therefore avoid statements that imply automation is exclusive to 4.0; instead, we characterize 5.0 as repurposing and re-governing automation.

Concurrently, parallel transformations are conceived in the domains of education and work. Education 5.0 has gained currency as a forward-looking framework for reimagining learning in the age of intelligent

* Correspondence to: Faculty of Law, Humanities and Social Sciences Department of Sociology, Walter Sisulu University, South Africa.
E-mail addresses: bchigbu@wsu.ac.za (B.I. Chigbu), lmakapela@wsu.ac.za (S.L. Makapela).

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machines and societal change (Alharbi, 2023; PwC/ASSOCHAM, 2024; Shahidi Hamedani et al., 2024). This framework builds on "Education 4.0" (the digital and personalized learning paradigm aligned with Industry 4.0) and extends the model by integrating AI and data-driven tools with humanistic pedagogies that emphasize creativity, emotional intelligence, and lifelong learning (Shahidi Hamedani et al., 2024; UNESCO, 2021b).

Likewise, the notion of Work 5.0 (or the "future of work 5.0") is emerging to describe an evolving labor landscape where technology augments human labor under improved conditions of equity, sustainability, and worker agency (IFOW, 2021; Plattform Industrie 4.0 4.0 4.0, 2025). Work 5.0 envisions a future of work that harnesses automation and digital platforms to enhance job quality, not degrade it, a future in which good jobs, social inclusion, and fair working conditions are pursued alongside productivity gains (Oeij et al., 2023; WEF, 2023).

These concepts, Industry 5.0, Education 5.0, and Work 5.0, have developed in parallel within their respective spheres. Each addresses critical issues posed by rapid technological advancement (especially AI and robotics), environmental sustainability imperatives, and shifting socio-economic needs in the post-pandemic era (Gola et al., 2025; Shahidi Hamedani et al., 2024). However, there is a growing recognition that these domains are deeply interdependent and that realizing their full potential requires an integrated approach (European Commission, 2021; Oeij et al., 2023; Shahidi Hamedani et al., 2024) (while acknowledging region-specific pathways shaped by governance capacity, financing, infrastructure, and labor institutions).

For example, Industry 5.0's success in creating human-centric and sustainable industries will depend on a workforce with new competencies – a need that Education 5.0 must fulfill through reimagined curricula and pedagogies (Grego et al., 2024; Supriya et al., 2024; Zabalawi and Kordahji, 2025). Similarly, the positive vision of Work 5.0 (e.g., "good work" in an AI-driven economy) hinges on both industrial innovation (creating new roles, augmenting jobs) and educational systems that prepare and continually upskill workers for these roles (IFOW, 2021; Plattform Industrie 4.0 4.0 4.0, 2025; WEF, 2023). Public policy and governance link all three: governments and international bodies are formulating strategies to guide industry innovation, reform education, and protect workers in tandem (UNESCO, 2021b; WEF, 2023). Conversely, advances in education and labor policy also shape and constrain industrial innovation trajectories; causality in our model is therefore reciprocal rather than unidirectional.

Existing literature has tended to discuss these paradigms in silos, leading to fragmented insights and limited cross-domain theorisation. This creates a conceptual gap: how do the 5.0 paradigms mutually constitute the future of work and education under conditions of industrial transformation? By addressing this gap, our paper contributes a novel theoretical synthesis that identifies cross-cutting mediators (e.g., skills development, AI ethics, sustainability) and highlights underexplored contradictions (such as the coexistence of human-centric rhetoric and persistent automation logics). Furthermore, the analysis intersects with research on open innovation dynamics. Foundational contributions (See: Chesbrough and Bogers, 2014; Yun et al., 2020) emphasize bounded rationality, culture, and complexity as drivers and constraints in collaborative innovation systems. These insights enrich the 5.0 debate by situating human-centric transitions within broader patterns of collective intelligence, platform-mediated exchange, and Small and medium-sized enterprises (SMEs) innovation practices.

This article addresses that gap by conducting an integrative literature review spanning the interdisciplinary intersections of Industry 5.0, Education 5.0, and Work 5.0. An integrative review is a distinctive research approach that synthesizes existing literature, including theoretical papers, empirical studies, and policy reports, to generate new understanding or frameworks for emerging topics (Torraco, 2016). This method suits our inquiry well because the "5.0" concepts are nascent and inherently interdisciplinary, involving technological, pedagogical, economic, and social dimensions.

By reviewing scholarship and reports across education technology, industrial engineering, management, labor economics, public policy, and sustainability science, we develop an integrative conceptual framework linking the three domains. We also draw on recent global data and policy developments (e.g., from the WEF, European Commission, and UNESCO) to analyze current trends and ensure relevance to ongoing debates. We deliberately focus on Industry 5.0, Education 5.0, and Work 5.0 because (i) they are the three 'systems of action' in which human-AI collaboration, skills formation, and socio-technical governance most directly co-evolve; (ii) they are the units at which major international frameworks and datasets exist (European Commission, 2021; UNESCO, 2021b; WEF, 2023) enabling rigorous cross-domain synthesis; and (iii) they form a tightly coupled, bidirectional triad-education builds capabilities for work and industry; industry sets technological trajectories and demand; and work institutions translate both into job quality and inclusion. Other "5.0" labels (e.g., Society 5.0, Government 5.0, Healthcare 5.0, Smart Cities 5.0) were excluded as co-equal domains to avoid scope creep and conceptual dilution; where relevant, we treat them as contextual environments or application sectors nested within the triad.

Human-centric: a design and governance orientation that centers human agency, dignity, capability development, safety, and meaningful participation in technology decisions and outcomes; this implies human-in-the-loop defaults and worker/learner voice. Sustainability: we use the tripartite definition (ecological, social, and economic) and align examples to relevant Sustainable Development Goals (SDGs); we use social sustainability to denote distributional equity, inclusion, job quality, and community well-being. AI integration: the purposeful incorporation of AI systems into educational, industrial, and work processes under augmentation-first principles (task allocation, oversight, transparency, and data governance), distinct from automation-only substitution. "Good work": used here to denote decent work/job quality with fair pay, security, autonomy, and worker voice, safe/healthy conditions, opportunities for learning and progression, and inclusion, and treated as a component of social sustainability throughout the manuscript.

The structure of the article follows the conventions of a scholarly review. We first present our inquiry's conceptual framework, illustrating how Industry 5.0, Education 5.0, and Work 5.0 connect and overlap. Next, we outline the integrative review methodology, including our search strategies and inclusion criteria. The heart of the article is organized around thematic findings that emerged from the literature synthesis: (1) human-machine collaboration and *AI integration* as a unifying theme of Industry 5.0 and Education 5.0, (2) skills development and curriculum alignment bridging education and the future of work, and (3) sustainability and social well-being as key goals across all domains. In each thematic sub-section, we integrate insights from multiple fields and highlight examples of initiatives or research. A discussion section then interprets these findings, examining implications for theory and practice. We consider how an interdisciplinary 5.0 paradigm can inform policy (for instance, aligning education reforms with industrial and labor strategies) and identify challenges and tensions (such as ethical considerations of AI in education and workplaces). Finally, we conclude with a summary of contributions, acknowledge the review's limitations, and suggest directions for future research at the nexus of industry, education, and work in the 5.0 era. This review synthesizes the fragmented 5.0 discourses by bringing together these perspectives and offers a novel conceptual lens for understanding their intersections, tensions, and shared imperatives. In doing so, it advances a new social contract for technology and human capital that is human-centric, inclusive, and sustainable. It also points to the limitations and contextual challenges that future empirical research must address.

2. Conceptual framework: linking industry 5.0, education 5.0, and work 5.0

This section defines constructs (nodes), relationships (edges), and mediators only; empirical illustrations, examples, and thematic findings are reserved for 4 (Findings). Industry 5.0, Education 5.0, and Work 5.0 intersect through the core principles of human-centric innovation and sustainability. Industry 5.0 contributes advanced technologies and a human-centric paradigm for production; Education 5.0 contributes skilled, adaptable talent and innovation capacity; and Work 5.0 ensures that technological change translates into "good work" and social well-being. We explicitly conceptualize these domains as overlapping and mutually constitutive, not hierarchical; influences run in bidirectional-evolutionary feedback loops rather than a linear pipeline. The overlap highlights a shared commitment to human-machine collaboration, digital transformation aligned with human values, and SDGs across all three domains. Fig. 1 depicts non-hierarchical, bidirectionally linked relationships among the three "5.0". Arrows indicate co-evolutionary feedback loops; human-centricity and sustainability are shared principles across all nodes. Accordingly, we treat skills development not as a fourth domain but as an Education 5.0 function that mediates the Education-Work interface and supports Industry 5.0 adoption (e.g., curriculum co-design, credentialing, and lifelong learning). For clarity of scope, we model these three as mutually constitutive, overlapping domains rather than a hierarchy; excluded "5.0" labels are handled as cross-cutting contexts that feed into these nodes rather than additional nodes in the diagram.

Yet, the integrative map also exposes important tensions that require a critical lens. First, "human-centric" innovation can slip into techno-optimism if worker agency, teacher autonomy, and community voice are not structurally protected; second, the 5.0 discourse originates mainly in the Global North, raising questions about transferability to low-resource systems and the risk of norm diffusion without capacity (policy, funding, infrastructure). Third, the overlapping aims of productivity, inclusion, and sustainability can conflict in practice, for example, when short-term efficiency pressures undermine job quality or when green process upgrades impose costs that smaller firms or public schools cannot absorb. Therefore, a realistic 5.0 framework must acknowledge governance capacity, distributional effects, accountability mechanisms, and aspirational goals.

We adopt *human-centric innovation* and *sustainability* as shared organizing principles across Industry 5.0, Education 5.0, and Work 5.0. Each domain brings a distinct focus: Industry 5.0 is primarily about technological innovation with a deliberate shift toward human-centric and sustainable production (Adel, 2022; European Commission, 2021; Grybauskas and Cárdenas-Rubio, 2024; Shahidi Hamedani et al., 2024). Education 5.0 centers on innovation in teaching and learning to empower individuals with the skills and values needed in a technology-rich, sustainable society (PwC/ASSOCHAM, 2024). Work 5.0 concentrates on innovation in work organization and labor policy to ensure inclusive, meaningful employment in the face of automation and digital platforms. Despite these different entry points, the domains overlap significantly in objectives and must operate in tandem. In the overlapping region of Fig. 1, we delineate cross-domain constructs (e.g., human-AI collaboration, skills pathways, sustainability linkages) while deferring operational themes and evidence to 4 (Findings). We treat these as cross-cutting principles rather than a base layer to avoid implying hierarchy.

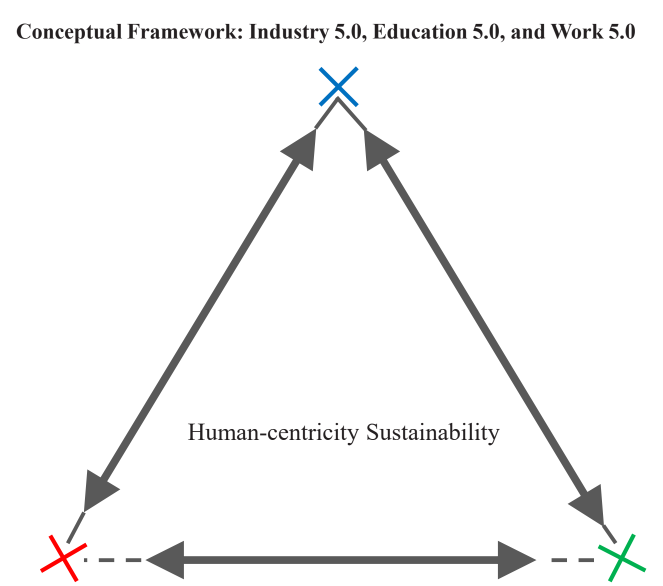

Fig. 1. Triadic co-evolution diagram with bidirectional arrows of Industry 5.0, Education 5.0, and Work 5.0. Source: Developed by the authors based on the synthesis of European Commission (2021); Rejeb et al# (2025); Shahidi Hamedani et al# (2024); UNESCO (2021); WEF (2023).

Zimbabwean HEIs report connectivity constraints, budget limits, resistance to change, and training gaps; UNESCO's policy guidance flags systemic risks (privacy, equity, procurement, teacher capacity). These factors condition outcomes in Education 5.0 and caution against universal claims (UNESCO, 2021a; Zishiri et al., 2024).

First, all three paradigms embrace human-machine collaboration instead of a purely automation-centric model. Industry 5.0 envisions factories where collaborative robots ("cobots") and AI systems assist human workers, augmenting their capabilities and reducing drudgery on the shop floor (Adel, 2022; Chen et al., 2024; Chigbu and Nekhwevha, 2022). Education 5.0 likewise explores AI tutors, personalized learning software, and immersive technologies (VR/AR) as tools to enhance, not replace, human teaching and learning, keeping teachers and students "in the loop" (Shahidi Hamedani et al., 2024). Work 5.0 foresees jobs redesigned such that humans work alongside AI decision-support systems or robotic assistants, which requires co-designing work tasks to achieve optimal human-machine synergy. Essentially, the 5.0 vision in all domains rejects a purely automated, machine-driven future. It emphasizes augmentative collaboration, where technology is developed *with* and *for* people. This theme reflects a broader shift from the automation ethos of Industry 4.0 (Caiado et al., 2024; Leng et al., 2022) to an "augmentation" ethos in 5.0 rather than aiming to remove humans from processes. The goal is to optimize human-machine interactions to capitalize on human creativity, flexibility, and judgment while maintaining efficiency (Adel, 2022). For example, in an Industry 5.0 smart factory, an AI system might detect an anomaly in production data. However, rather than automatically shutting down a process, it alerts a human engineer, who then uses insight and creativity to decide on a synergistic solution between machine speed and human judgment. Early case evidence suggests that such approaches can yield benefits: pilot implementations of Industry 5.0 report increased worker satisfaction when employees feel technology supports them rather than monitors or replaces them (Anand et al., 2023; Trstenjak et al., 2025). Recent studies on human-centric Industry 5.0 found that productivity, efficiency, and job satisfaction all rise when humans and smart machines collaborate as partners (Adel, 2022; Anand et al., 2023; Chen et al., 2024; Grybauskas and Cárdenas-Rubio, 2024; Leng et al., 2022). Additionally, incorporating worker feedback into the design of AI systems (for example, co-designing an AI-based scheduling tool with employee input) can improve user adoption and trust in these technologies. In sum, human-machine collaboration represents a unifying principle of the 5.0 paradigms, underscoring that the endpoint of technological innovation is to empower humans, not supplant them (Adel, 2022; Bucci et al., 2025; Chen et al., 2024; European Commission, 2021; Gaffinet et al., 2025; Tóth et al., 2023; Wang et al., 2024).

Second, the domains converge on the importance of skills development and curriculum alignment for the future of work. A prominent challenge of the 5.0 era is ensuring that education and training systems keep pace with the rapidly evolving skill requirements driven by AI, automation, and sustainability transitions. Industry 5.0 workplaces will demand not only advanced technical skills (e.g., robotics programming, data analytics) (Maddikunta et al., 2022) but also transversal skills like problem-solving, creativity, adaptability, and collaborative competencies sometimes called "21st-century skills" which Education 5.0 seeks to cultivate (PwC/ASSOCHAM, 2024; Shahidi Hamedani et al., 2024; Tiron-Tudor et al., 2025). Likewise, Work 5.0 emphasizes continuous learning and reskilling as fundamental to helping workers thrive alongside intelligent machines (Oeij et al., 2023). Our review found widespread acknowledgment of a growing skills gap: the workforce is not fully prepared with the skills needed for emerging Industry 5.0 jobs, nor are current educational curricula sufficiently aligned to these future needs. Recent global data underscore this concern. For instance, companies report that skills shortages and an inability to attract qualified talent are among the key barriers to adopting new technologies, highlighting a clear need for training and reskilling across industries (Chigbu and Makapela, 2025; WEF, 2023). According to the WEF's *Future of Jobs Report 2023*, six in ten workers will require upskilling or reskilling by 2027. However, only about half of workers have access to adequate training opportunities (WEF, 2023). This urgent need has led to calls for closer integration between educational institutions, industry, and policymakers to "bridge the gap" between education and the future of work. Examples of this integration include updating school and university curricula in partnership with industry to include competencies for the digital and green economy, expanding vocational and on-the-job training programs in alignment with Work 5.0 principles, and recognizing new forms of credentials (such as micro-credentials or industry certifications) that validate the skills required in 5.0 workplaces (Supriya et al., 2024; Tiron-Tudor et al., 2025; WEF, 2023; Zabalawi and Kordahji, 2025). Evidence from chemical engineering shows how curricula can explicitly map Education 5.0 to Industry 5.0 skills (Galatro and Chakraborty, 2025). In summary, the Education 5.0 and Work 5.0 agendas are intrinsically linked through the common goal of talent development for a human-centric, AI-augmented economy. Addressing the skills and education gap is considered a prerequisite for realizing the promises of Industry 5.0 innovation in a socially inclusive way.

Third, sustainability and social well-being emerge as shared priorities that cut across Industry 5.0, Education 5.0, and Work 5.0 strategies. Unlike the purely efficiency-driven mindset of some earlier industrial paradigms, the new 5.0 frameworks explicitly incorporate goals of environmental sustainability and human well-being as measures of success (European Commission, 2021, 2025; Rejeb et al., 2025). Industry 5.0, for example, calls on manufacturers to adopt processes and technologies that respect environmental limits (e.g., lowering carbon emissions, reducing waste, using renewable energy), thus positioning the industry as a key contributor to achieving societal sustainability goals (European Commission, 2021; Leng et al., 2022). The European Commission's vision for Industry 5.0 notably frames industry as a *"provider of prosperity"* that must operate within planetary boundaries and deliver value to society, not just to shareholders (European Commission, 2021). Similarly, Education 5.0 emphasizes education for sustainable development, cultivating learners who have the knowledge, values, and agency to address global challenges such as climate change and inequality. This includes integrating the SDGs into curricula and promoting interdisciplinary learning about sustainability issues (UNESCO, 2021b). Work 5.0 likewise extends beyond traditional labor metrics to consider job quality, equity, and community well-being. It encourages organizational and policy innovations (e.g., green jobs programs, fair work initiatives, diversity and inclusion measures) that ensure the future of work contributes to broader social well-being and does not exacerbate inequalities. In all three domains, there is a normative shift toward viewing technological advancement as a means to *human* and *planetary* ends rather than an end in itself. This alignment is evidenced by common references to frameworks like the United Nations SDGs, "Society 5.0" (the Japanese concept of a super-smart society that balances economic advancement with social good), and various "Tech for Good" initiatives. The convergence on sustainability and well-being also means that progress in one domain reinforces the others: for instance, educating students in sustainable practices (Education 5.0) produces a workforce capable of implementing sustainable industry innovations (Industry 5.0) and advancing green jobs (Work 5.0). All three together envision a future in which economic growth, human development, and environmental stewardship advance hand-in-hand.

These three thematic areas, human-machine collaboration, skills/curriculum alignment, and sustainability/social well-being, form the core of an integrated Industry–Education–Work 5.0 paradigm. In the Findings section (4), we delve deeper into each theme, drawing on literature and examples to illustrate how the intersections manifest in practice. By exploring these themes, we can better understand how coordinated technological changes, human capital development, and organizational values might collectively shape a more human-centric and sustainable future.

3. Methodology

3.1. Research design

This study employed an integrative literature review methodology to gather and synthesize knowledge from diverse sources related to Industry 5.0, Education 5.0, and Work 5.0. An integrative review is a form of research that reviews, critiques, and synthesizes representative literature on a topic in an integrated way so that new frameworks and perspectives emerge. This method allows the inclusion of various sources (e.g., theoretical articles, empirical studies, policy reports, and reputable industry publications). It is beneficial for examining emerging, interdisciplinary topics not yet dominated by a large, coherent body of empirical research (Torraco, 2016). Given that "5.0" concepts have only gained prominence in the past few years (Leng et al., 2022) and span multiple domains, an integrative approach was appropriate to capture the state of the art and to connect insights across fields. We specified our scope, data sources, search strings, screening steps, and inclusion/exclusion criteria a priori to enhance transparency and report them below.

3.2. Search strategy

We searched Scopus, Web of Science Core Collection, and Google Scholar. Across these databases and portals, approximately 200 records were retrieved, with the majority originating from Scopus and Google Scholar. Exact database-level breakdowns were not logged at the initial stage; however, a transparent PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flow (Fig. 2) documents the screening process from identification to inclusion. In addition, we systematically reviewed authoritative grey-literature portals: European Commission (Industry 5.0), United Nations Educational, Scientific and Cultural Organization (UNESCO) (Futures of Education/AI in education), WEF (Future of Jobs/Skills), International Labour Organization (ILO) (future of work/decent work), and Organisation for Economic Cooperation and Development (OECD) (skills/lifelong learning). The primary date window was January 2020–September 2025 (earlier seminal works included for conceptual background). Language was limited to English; eligible document types included peer-reviewed journal articles, scholarly conference papers, books/chapters, and policy/strategy reports from the listed organizations. Database queries were run in Title/Abstract/Keywords using Boolean operators. Representative strings are provided in Table 2. We performed field scoping, title/abstract screening, full-text review, backward/forward citation chasing (snowballing), targeted grey-literature retrieval via site search/portal taxonomies, and a recency filter prioritizing 2020–2025 (with pre-2020 works included when conceptually foundational). Table 1, databases and sources searched (structured list of academic databases and institutional portals) appears below.

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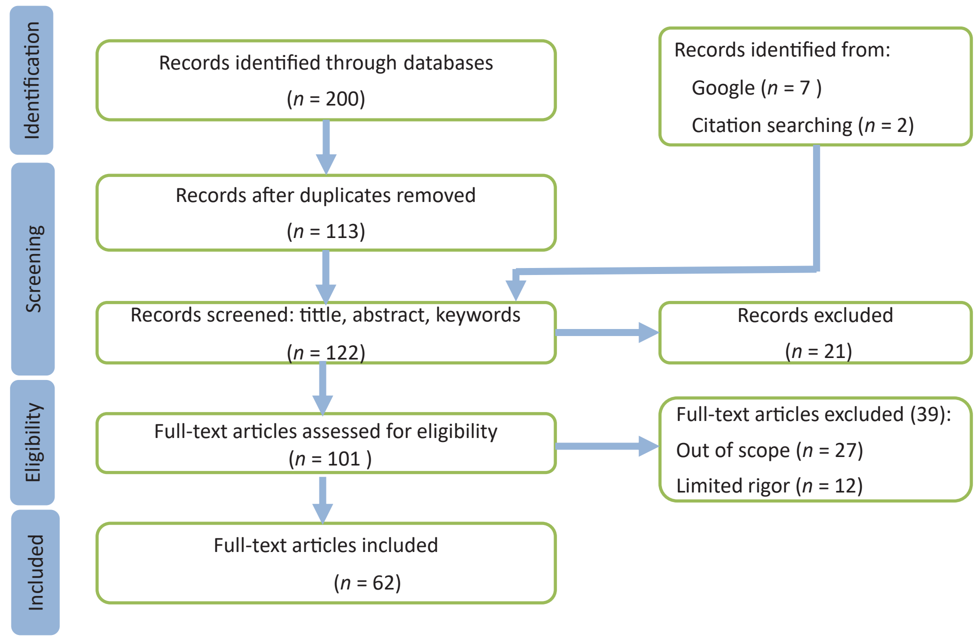

Fig. 2. The PRISMA flow diagram adapted for an integrative literature review of Industry 5.0, Education 5.0, and Work 5.0 literature illustrates source identification, screening, eligibility, and inclusion (n = 62). Source: Authors.

Table 1
Databases and sources searched.

Source	Coverage/focus	Notes
Scopus	Multidisciplinary scholarly literature	Title/Abstract/Keywords searched; 2020–2025 emphasis
Web of Science Core Collection	Multidisciplinary scholarly literature	Title/Abstract/Keywords searched; 2020–2025 emphasis
Google Scholar	Broad academic & preprints	First 200 hits per query screened for relevance/date
European Commission	Industry 5.0 policy & communiqués	Grey literature (official communications/reports)
UNESCO	Futures of education; AI in education	Grey literature (global education reports/guidance)
WEF	Future of jobs/skills	Grey literature (white papers, reports)
ILO	Future of work; decent work	Grey literature (conventions, reports)
OECD	Skills, lifelong learning, adult training	Grey literature (policy papers, data)

Source: Authors

3.2.1. Selection and inclusion

Over 200 sources were initially identified but 62 sources were finally included in synthesis (see PRISMA, Fig. 2). Inclusion criteria (applied at title/abstract/full-text): (i) explicitly define or theorize *Industry 5.0*, *Education 5.0*, or *Work 5.0*; (ii) empirically examine implementations/impacts of these paradigms; or (iii) constitute authoritative policy/strategy documents from EC, UNESCO, WEF, ILO, or OECD. We included sources published primarily from 2020 to 2025 (with seminal pre-2020 works for background). We excluded duplicates; non-scholarly media without substantive analysis; Industry 4.0-only records without an explicit bridge to 5.0; unreferenced opinion pieces; and works lacking sufficient methodological detail for interpretation. Key information (domain, method, geography, sector, outcomes) and themes (e.g., human-centric design, skills/upskilling, AI in education/work, sustainability, worker well-being) were extracted into a structured matrix to support cross-domain analysis. Table 3 summarizes the inclusion/exclusion criteria at a glance.

The selection process followed an adapted PRISMA flow to ensure transparency and rigor in source inclusion. The flowchart (see Fig. 2) summarizes the number of records identified, screened, assessed for eligibility, and included in the review. Ultimately, 62 full-text sources met the criteria and were incorporated into the final synthesis.

To strengthen reliability, we also conducted a structured quality appraisal. For empirical studies, we adapted criteria from the Mixed Methods Appraisal Tool and Critical Appraisal Skills Programme checklists, focusing on clarity of aims, methodological rigor, and transparency of data/analysis. Credibility was appraised for policy and strategy reports based on issuing body (e.g., EC, UNESCO, ILO, WEF) and methodological disclosure. Findings from high-quality empirical studies were weighted more heavily in synthesis (e.g., triangulated cobot/AI pilot evaluations, large-sample surveys). At the same time, policy sources were used primarily to contextualize agendas and highlight normative commitments.

Table 2
Example Boolean queries (Title/Abstract/Keywords).

Theme	Example Boolean string
Core concept	"Industry 5.0" OR "fifth industrial revolution"
Education interface	("Education 5.0" OR "future of education") AND ("AI" OR "skills" OR "curriculum" OR "ethics")
Work interface	("Work 5.0" OR "future of work 5.0") AND ("job quality" OR "decent work" OR "reskilling" OR "AI")
Cross-cutting	("Industry 5.0" AND sustainability) OR ("Industry 5.0" AND "human-centric")
Bridging terms	("Industry 5.0" AND education) OR ("Industry 5.0" AND "future of work")

Source: Authors

Table 3
Inclusion and exclusion criteria at a glance.

Category	Criteria
Include	Defines/theorizes I5.0/E5.0/W5.0; empirical studies on 5.0 applications; EC/UNESCO/WEF/ILO/OECD policy reports; 2020–2025; English
Exclude	Duplicates; non-scholarly media; 4.0-only without bridge to 5.0; unreferenced opinion; insufficient methodological detail

Source: Authors

Table 4
Regional variability in 5.0 implementation.

Region	Enablers	Binding constraints	Implications for 5.0
EU	Regulatory capacity; green funds; social dialogue	SME retrofit costs; standards fragmentation	Augmentation focus; ESG alignment; skills compacts
Japan	Society 5.0 policy; ageing-care demand	Labor shortages; conservative adoption	Care robotics; ethics-centric Education 5.0; human-AI teaming
Sub-Saharan Africa	Youthful demographics; entrepreneurship policy	Connectivity; devices; teacher PD; fiscal space	Phased AI integration; open resources; Technical Vocational Education and Training (TVET)-industry hubs
South/Southeast Asia	Large skilling missions; export sectors	Quality assurance; rural-urban divides	Scaled micro-credentials; targeted regional investment
Latin America & Caribbean	Active social policy; cluster potential	Informality; platform precarity; fiscal volatility	Formalization; digital rights; place-based Industry 5.0
Small island/fragile states	Tight communities; niche sectors	Climate risk; import dependence; brain drain	Resilience-first Work 5.0; regional consortia for skills

Source: Authors

3.3. *Integrative synthesis*

We employed an integrative synthesis approach to integrate findings. Rather than presenting separate systematic reviews of each domain, we organized the evidence around cross-cutting themes that answered our guiding question: ***How do Industry 5.0, Education 5.0, and Work 5.0 collectively inform a human-centric, sustainable future?*** We iteratively developed the themes presented in the Findings section by examining where multiple sources overlapped in content. For example, many sources across industry and labor domains emphasized human-machine *collaboration* (rather than a replacement) and upskilling; this coalesced into a theme of human-AI collaboration and skills. During this synthesis, we also explicitly constructed the conceptual framework (Fig. 1) to visualize the relationships identified. The framework was refined by verifying each link with literature (e.g., confirming that scholars or policymakers have linked education reforms with Industry 5.0 requirements (Luna et al., 2024; Oeij et al., 2023) or linking Industry 5.0 and work quality improvements). All synthesis steps were traceable to the screening matrix described above, ensuring consistency between inclusion decisions and thematic coding.

All statements or direct claims are supported by citations to the literature, using real scholarly or authoritative sources. The reference list was compiled per APA guidelines, including each work's author name (or organization), year, title, and source details. Where we include visuals (such as the conceptual Fig. 1), we have provided appropriate attributions or created the figure based on synthesized concepts from the literature. The subsequent Findings section operationalizes the framework via three thematic analyses and presents empirical exemplars; no results are previewed in the framework.

4. Findings

Through the integrative analysis, we identified several major themes that represent the intersection of Industry 5.0, Education 5.0, and Work 5.0. In many settings, these themes recur across scholarly discussions, policy documents, and practical case studies; however, trajectories and feasibility vary by region and development context. We present three key thematic areas:

- Human-Machine Collaboration and AI Integration (the shift from automation to *augmentation* and the implications for industry workflows, educational practices, and job roles in the future of work).
- Education 5.0-led skills development & lifelong learning (mediating the Education and Work interface for Industry 5.0 adoption)for the Future of Work (addressing the digital skills gap, needed reforms in curricula and training programs, and new models of lifelong learning to prepare a workforce for the 5.0 era).
- Sustainability and Social Well-Being in 5.0 Strategies (how environmental sustainability and social responsibility are embedded as common goals in each domain, and how industry, education, and work initiatives reinforce one another toward these goals).

Each thematic section below integrates insights from the literature and provides examples or data where applicable.

4.1. *Human–machine collaboration and AI integration*

Beyond cobots, the 5.0 technology stack spans complementary systems: digital twins and simulation for operator-in-the-loop optimisation and predictive maintenance; IoT/edge/5G sensing for real-time visibility; AI decision-support (e.g., vision, forecasting, scheduling) under human oversight; and VR/AR for training, remote assistance, and safety-critical rehearsal. We therefore discuss human-machine teaming across this broader stack rather than robotics alone.

A defining characteristic of the 5.0 paradigm shift is the reconceptualization of the relationship between humans and machines. Under the earlier Industry 4.0 narrative, there was often an implicit goal of maximizing automation, i.e., having machines perform as many tasks as possible with minimal human intervention. In contrast, Industry 5.0 explicitly emphasizes collaboration between humans and machines. This theme of human-machine collaboration or *"augmentation"* also permeates Education 5.0 and Work 5.0 since AI and digital tools are increasingly present in classrooms and workplaces. However, they are being leveraged in ways that keep humans in the loop. The overarching idea is that technology should complement human strengths rather than replace human workers or teachers. Accordingly, we do not treat automation as receding in 5.0; instead, automation is redesigned to serve human capabilities through human-in-the-loop control, transparency, and co-design. In this section, "AI integration" follows the operational definition above (augmentation-first, human-in-the-loop, governed by transparency and data ethics).

Across early deployments, human-AI/cobot systems show measurable benefits when co-designed with operators: SMEs report improved ergonomics and higher perceived autonomy with stable or increased throughput; AI-vision with human override reduces false stops while maintaining safety; and participatory Human-machine interface/scheduling design increases trust and adoption (European Commission, 2021; Grego et al., 2024; Grybauskas and Cárdenas-Rubio, 2024; Rejeb et al., 2025).

- o SME assembly with cobots: improved ergonomics/autonomy; stable or output (Grybauskas and Cárdenas-Rubio, 2024; Rejeb et al., 2025).
- o AI-vision + human override: false stops; safety preserved (European Commission, 2021; Grego et al., 2024).
- o Co-designed Human-machine interface/scheduling: higher trust and sustained use (Rejeb et al., 2025).

In the industrial context, *cobots and* digital twins, IoT/edge connectivity, and AI decision-support exemplify this approach. Instead of large industrial robots fenced off from people (as in traditional automation), cobots are designed to work side-by-side with human operators, assisting with tasks and responding to human actions. They might handle heavy lifting or precision positioning while a human worker performs the skilled assembly or quality control. The European Commission notes that optimizing such "human-machine interactions" allows the industry to capitalize on humans' unique creativity and problem-solving abilities while maintaining high efficiency through automation (European Commission, 2021). For example, in a smart factory setting, AI-based vision systems might detect anomalies on a production line. However, rather than automatically shutting the line down, the system could alert a human technician, who then uses expertise to decide corrective action, combining machine speed and consistency with human judgment (Maddikunta et al., 2022). Early implementations of Industry 5.0 report positive outcomes from this collaborative approach. In early studies, workers who interacted with assistive robots described higher job satisfaction, feeling that technology was a supportive "colleague" rather than a competitor or surveillance tool (Anand et al., 2023; Maddikunta et al., 2022). Indeed, recent reviews concluded that Industry 5.0's human-centric approach tends to "empower humans and improve human-machine interaction in dynamic industrial systems," ultimately enhancing worker well-being without sacrificing productivity (Alves et al., 2023; Leng et al., 2022; Vuori et al., 2025). While early studies are promising, a balanced view is essential. Augmentation can also introduce *new* risks: (i) creeping deskilling when decision authority quietly migrates from workers to algorithms; (ii) intensification and cognitive overload as humans supervise multiple systems; and (iii) surveillance-driven "collaboration" that can erode autonomy and trust. Empirical results remain mixed across sectors; impacts depend on participatory design, explainability, ergonomics, and negotiated workplace safeguards (e.g., co-determination, data governance, right-to-disconnect).

In Education 5.0, a similar philosophy guides the integration of AI and digital tools. Rather than aiming to automate teaching or replace educators, Education 5.0 initiatives focus on using technology to *enhance* human teaching and personalize student learning. For instance, AI-driven tutoring systems can provide students with adaptive feedback and extra practice on basic skills, freeing up teachers' time to focus on higher-level mentoring and social-emotional support. Virtual and augmented reality can immerse students in interactive learning experiences that would be impossible to create otherwise, requiring teachers to guide reflection and context. The phrase "AI in education" in this paradigm often comes with the caveat that AI is a tool to *support* teachers and learners, not a substitute for the human elements of education (such as empathy, inspiration, and ethical judgment) (UNESCO, 2021b). Researchers emphasize that keeping teachers and students "in the loop" is crucial – meaning teachers remain central to orchestrating AI-enhanced learning, and students are taught to work effectively with AI tools as partners in the learning process (Seeling et al., 2022; Vuori et al., 2025). This human-centric approach in Education 5.0 aligns with UNESCO's guidance on AI in education, which asserts that AI should augment teachers' capacities and help personalize learning. At the same time, educators uphold the pedagogical and ethical foundations of teaching (UNESCO, 2021b). For example, a classroom might use an AI tutor for mathematics practice. However, the teacher monitors students' progress via the AI dashboard and intervenes with human insight when a student is struggling with conceptual understanding or motivation, leveraging the best of both AI and human abilities (Seeling et al., 2022). Beyond tutoring, implementations include learning analytics/dashboards, AI-assisted assessment, and VR/AR labs; emerging "digital twins of learning" mirror progression and workload to inform human judgment.

Dynamic Capabilities 5.0 multi-case evaluation prioritises collaborative ecosystems and digital readiness over people-centric innovation as drivers of resilience (Mohammed et al., 2025). Lean-5.0 cases report benefits but only with targeted capability building and change management (Fani et al., 2024). Human Digital Twin (HDT) surveys highlight unresolved standards, privacy, and behavioural-modeling gaps that limit the generalisability of shop-floor gains (Bucci et al., 2025; Gaffinet et al., 2025; Wang et al., 2024).

In the realm of Work 5.0, the collaborative paradigm is often encapsulated in the idea of "AI as a colleague, not just a tool (Anthuvan and Maheshwari, 2025; Ghugar et al., 2025; Krause et al., 2024; Plattform Industrie 4.0 4.0 4.0, 2025; Tóth et al., 2023)." The future workplace envisioned by Work 5.0 is one where humans and AI systems function as teams, each contributing to what they do best. AI or robots might handle routine and repetitive tasks, while humans focus on complex decision-making, creative problem-solving, interpersonal communication, and other areas where human skills excel. A concrete example is in healthcare: AI algorithms can rapidly analyze medical images or predict patient risks, but doctors and nurses collaborate with these AI systems by validating suggestions, providing the empathetic human touch in patient care, and making final judgment calls. This kind of human-AI teaming is already being piloted in some hospitals (Rabhi et al., 2025). Similarly, in corporate settings, "virtual colleagues" (AI assistants) are starting to assist with project management or data analysis tasks, working alongside human employees. Realizing the full potential of such arrangements will require thoughtful *co-design* of work processes – workflows must be redesigned so that AI and employees can seamlessly hand off tasks and support each other. Studies on human-AI collaboration in workplaces indicate that factors like trust, transparency, and worker involvement in AI adoption are key to success (Anthuvan and Maheshwari, 2025; Rabhi et al., 2025; Vuori et al., 2025). Workers are more likely to embrace AI tools if they understand how they function (explainability) and if they have had a say in how the tools are implemented to help their work (Anthuvan and Maheshwari, 2025). Thus, when introducing advanced technologies, Work 5.0 emphasizes change management practices, including employee training and feedback loops. This approach echoes the Work 5.0 principle that technology implementation should be *worker-centered*, aiming to *"optimally combine the strengths of humans and machines"*. When done right, such integration can lead to not only higher productivity but also improved job quality – for instance, reducing hazardous or tedious aspects of jobs and increasing the time workers spend on creative, meaningful tasks (Chen et al., 2024; Rabhi et al., 2025; Rejeb et al., 2025). Operationally, this spans AI-assisted scheduling and rostering, safety analytics from IoT wearables, and AR-guided procedures, implemented with negotiated guardrails on data use, autonomy, and worker voice.

Human-machine collaboration is a cornerstone of Industry 5.0, Education 5.0, and Work 5.0. All three domains are moving away from the notion of automation for its own sake and toward a vision of *symbiotic* relationships between humans and intelligent systems, wherein each amplifies the capabilities of the other. This theme holds significant promise: it suggests that the "fifth revolution" need not repeat past patterns of technological upheaval displacing humans. However, it can instead forge a future where technology elevates human work and learning to new heights. Accordingly, we treat human-AI teaming not as an inherent good but as *conditionally beneficial*, contingent on institutional arrangements that prevent substitution-by-stealth, protect privacy, and embed worker/teacher voice in design and deployment.

4.2. Education 5.0-led skills development and lifelong learning (mediating the Education-Work interface for Industry 5.0 adoption)

Another major intersection of the 5.0 domains lies in how they address the skills and education challenges of the coming decades. As industries transform and new technologies emerge, the skills required in the workforce are changing rapidly. Industry 5.0 and Work 5.0 highlight the need for workers who are not only technically proficient in working with advanced technologies (automation, data analytics, AI, etc.) but also adaptable, creative, and equipped with strong social and cognitive skills. At the same time, Education 5.0 is fundamentally about reorienting education systems to prepare individuals for this future – developing talent that can thrive in and shape the new landscape of work. The critical focal point is the alignment (or misalignment) between what education produces and what the future job market needs.

The pilot studies cited below show benefits in some settings, but findings are mixed and context-dependent; effects hinge on task design, worker training, and governance (Bucci et al., 2025; Fani et al., 2024; Mohammed et al., 2025; Tóth et al., 2023; UNESCO, 2021a; Wang et al., 2024; Zishiri et al., 2024).

Evaluations and pilots indicate concrete outcomes: AI-assisted tutoring increases practice completion while enabling teachers to reallocate time; university–industry "5.0" hubs produce job-relevant projects/placements; and employer-led reskilling improves internal mobility (Oeij et al., 2023; Plattform Industrie 4.0 4.0 4.0, 2025; Supriya et al., 2024; UNESCO, 2021b; WEF, 2023; Zabalawi and Kordahji, 2025).

- AI tutors in classrooms: practice completion; teacher time reallocated (UNESCO, 2021b).
- University-industry hubs ("5.0 academies"): curriculum co-design; authentic deliverables/placements (Luna et al., 2024; Plattform Industrie 4.0 4.0 4.0, 2025; Zabalawi and Kordahji, 2025).
- Employer-led reskilling: internal mobility/retention vs external hiring (Oeij et al., 2023; WEF, 2023).

A prominent concern driving Education 5.0 is the widely reported "skills gap" or "skills mismatch." Employers in many sectors have voiced that they struggle to find workers with the right mix of skills for modern roles, primarily as automation shifts the demand toward higher-order skills (ILO, 2022a; OECD, 2023; Tiron-Tudor et al., 2025). For example, while demand for basic cognitive and manual skills might decline with AI automation, demand is rising for complex problem-solving, critical thinking, digital literacy, and the ability to work alongside AI skills that are not traditionally taught in schools to a sufficient extent (Vuori et al., 2025; WEF, 2023). The WEF notes that analytical thinking and creative thinking are expected to remain the most important skills for workers, even as technical skills in AI and IT grow in prominence (WEF, 2023). However, many education systems are only beginning to integrate such skills into curricula. According to the same WEF report, employers estimate that, on average, 44 % of workers' core skills will need to be updated by 2027, reflecting an accelerating pace of change. This puts enormous pressure on formal education and lifelong learning mechanisms to adapt. The report also found that six in ten workers will require additional training in the next five years, but only about half of employees currently have access to sufficient training opportunities, a gap that risks leaving millions of workers behind if not addressed (ILO, 2022a; WEF, 2023).

Education 5.0 thus calls for bold changes in educational content and delivery. This includes updating curricula at all levels (Kindergarten through 12th grade, higher education, vocational training) to emphasize interdisciplinary learning, digital fluency, and 21st-century competencies (European Commission, 2021; Tiron-Tudor et al., 2025; UNESCO, 2021b). For instance, some universities are beginning to implement "Education 5.0" strategies by introducing programs in AI ethics, human–computer interaction, and sustainability, ensuring students acquire knowledge (Seeling et al., 2022) in technology *and* its societal context. There is also a push for more project-based and experiential learning models that mirror real-world problem-solving and encourage innovation, a departure from rote learning (Rabhi et al., 2025; Reimers and Schleicher, 2020). Additionally, Education 5.0 advocates for closer partnerships between educational institutions and industry. Such collaboration can ensure curricula remain relevant to industry trends and provide students with hands-on experiences (through internships, apprenticeships, and co-op programs) that build job-ready skills. An example is the emergence of "Industry 5.0 academies" or university-industry hubs where companies co-design coursework or projects with educators, often focused on areas like smart manufacturing, robotics, or sustainable design (Luna et al., 2024; Plattform Industrie 4.0 4.0 4.0, 2025). These initiatives reflect the understanding that education and work must co-evolve.

Work 5.0, for its part, emphasizes the need for continuous upskilling and reskilling of the existing workforce. In a Work 5.0 scenario, learning is not confined to the early years of life but becomes a lifelong endeavor supported by employers and society (Anthuvan and Maheshwari, 2025; ILO, 2022a). Companies implementing Work 5.0 principles invest in training programs to help their employees acquire new skills in emerging technologies rather than resorting solely to hiring new talent. Policies are also being considered; for example, some governments are exploring individual learning accounts or training subsidies as part of a broader employment strategy to facilitate lifelong learning (OECD, 2023). The ILO and other bodies have highlighted that proactive workforce development is key to preventing displacement and inequality in the face of automation (ILO, 2022a). In short, bridging the gap between education and the future of work involves reforming formal education (Education 5.0's domain) and creating robust systems for adult learning and career transitions (Work 5.0's domain). It is an inherently interdisciplinary challenge that requires educators, industry leaders, and policymakers to collaborate (Hashim et al., 2024; Reimers and Schleicher, 2020; UNESCO, 2021b).

However, the path from rhetoric to reality is uneven. Resource-constrained systems face teacher workload and professional-development deficits; infrastructure gaps (devices, connectivity) limit the feasibility of AI-enabled pedagogy; and the rapid proliferation of micro-credentials risks fragmenting quality assurance and exacerbating credentialism. Market-led skill agendas can also sideline civic, ethical, and foundational learning if not counter-balanced by public interest goals. Without sustained public investment and governance capacity, Education 5.0 and Work 5.0 may reproduce inequalities rather than close them, particularly across rural/urban divides and the Global South.

There are encouraging developments. The European Commission's *"Pact for Skills"* and various national strategies are aligning skill-building with Industry 5.0 and digital transition goals. In the education sphere, UNESCO's Futures of Education initiative calls for a new social contract in education that would ensure "education, knowledge, and innovation" are mobilized to create sustainable and just futures, implicitly recognizing the need for education systems to keep people employable and capable of driving positive change (UNESCO, 2021b). The Education 5.0 concept has also been explicitly taken up in some countries (for example, Zimbabwe's higher education policy labeled "Education 5.0" focuses on producing graduates with entrepreneurial and innovative skills to foster industrialization) (Government of Zimbabwe, 2020). These efforts illustrate that around the world, linking education reform with the future of work is becoming a policy priority. See Fig. 3 for a visual representation of how Education 5.0-led skills development and lifelong learning connect Education 5.0, Work 5.0, and Industry 5.0 in a mutual learning loop, highlighting their co-evolution toward shared societal and sustainability goals.

This infographic depicts the dynamic alignment between evolving industry demands, educational transformation, and workforce development under the 5.0 paradigm. It highlights closing the skills gap through digital literacy, interdisciplinary curricula, university-industry collaboration, and lifelong learning systems. In conclusion, Education 5.0-led skills development and lifelong learning (mediating the Education-Work interface for Industry 5.0 adoption) is a theme that underscores the tight coupling between the success of Industry 5.0 innovations and the readiness of the human workforce (Tiron-Tudor et al., 2025). The human capital dimension cannot be an afterthought; it must be planned in tandem. Industry 5.0 cannot reach its human-centric aspirations if education systems fail to produce human-centric innovators, and workers are left without pathways to upgrade their skills.

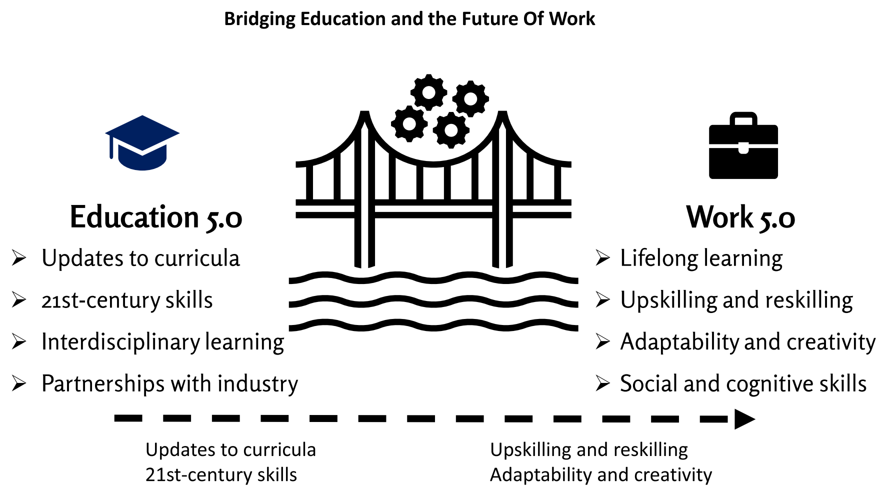

Fig. 3. *Skills Development and Curriculum Alignment for the Future of Work. Source: Authors.*

Conversely, the promise of Education 5.0 (empowering learners with cutting-edge skills and values) is ultimately measured by whether those learners thrive and lead in the changing world of Work 5.0. Thus, realising the 5.0 vision requires *sequenced and funded* reforms, curriculum co-design with industry and communities, robust teacher training, portable learning rights for adults, and equity-oriented financing, so that alignment does not devolve into short-cycle training that narrows opportunity. These findings parallel work on open and social innovation, where collective intelligence, platform participation, and business model adaptation enable systemic learning (Bogers et al., 2018; Chesbrough and Bogers, 2014; Yun et al., 2020). Linking Education 5.0 hubs with open innovation ecosystems can strengthen curriculum responsiveness and expand opportunities for SMEs and communities.

4.3. Sustainability and social well-being as shared goals

We apply a tripartite sustainability lens (ecological, social, and economic) and align each explicitly to the SDGs: ecological (SDGs 12–13), social (SDGs 3, 4, 5, 10, 16), and economic (SDGs 8–9). In our synthesis, Industry 5.0 contributes via circular/low-carbon processes; Education 5.0 via *education for sustainable development* (ESD) and climate/AI ethics; and Work 5.0 via decent-work, protection, and just-transition measures, linking firm design, curricula, and labour institutions to measurable outcomes (United Nations, 2015). This reflects an important normative shift in defining success in each domain. Traditionally, Industry 4.0 initiatives were evaluated on economic metrics like productivity gains or output growth, educational reforms on academic achievement or graduation rates, and labor policies on job creation numbers or Gross Domestic Product (GDP) growth. In the 5.0 paradigm, however, objectives are explicitly broadened to include environmental sustainability and human well-being as key outcomes.

Across sectors, documented pilots demonstrate operational sustainability: manufacturers implementing circular/energy-optimised processes report waste and emissions reductions; education systems embed ESD modules and campus initiatives; and employers trial working-time reforms with stable output and higher reported well-being (European Commission, 2021; Gola et al., 2025; IFOW, 2021; UNESCO, 2021b; WEF, 2023).

Empirical exemplars and evidence:

- o Circular/low-carbon manufacturing pilots: reduced material waste and energy intensity (Gola et al., 2025).
- o ESD-aligned curricula and green campuses: measurable changes in student projects and campus operations.
- o Working-time reforms (e.g., four-day week/right-to-disconnect): improved well-being gains with sometimes stable, sometimes neutral, and occasionally negative productivity effects in pilots; outcomes depend on sectoral context, job design, and social dialogue (Fani et al., 2024; IFOW, 2021; Mohammed et al., 2025; WEF, 2023).
- o Validation plan: Subject the Venn model to expert (Delphi) appraisal and target pilots with preregistered well-being/inclusion/productivity metrics across regions; revise the model accordingly.

The European Commission's Industry 5.0 framework explicitly calls for industry to *"achieve societal goals beyond jobs and growth"*, becoming *"a resilient provider of prosperity"* that respects the boundaries of our planet and places worker well-being at the center of production (European Commission, 2021). This is a striking evolution from earlier industrial paradigms, essentially a call for industry to contribute to solutions for climate change and inequality rather than being viewed solely as a source of those problems. In practical terms, Industry 5.0 encourages the adoption of sustainable technologies (Leng et al., 2022; Piccarozzi et al., 2024; Xu et al., 2021): for example, using circular economy principles in production (to minimize waste and reuse materials), adopting clean energy, and designing products that have positive societal impacts (Atif, 2023). It also means prioritizing *human-centric* metrics such as worker safety, health, and satisfaction in evaluating industrial performance. Early adopters of Industry 5.0 have reported pursuing initiatives like carbon-neutral factories, "lights-off" production that saves energy, and manufacturing assistive devices for social needs, aligning business strategy with sustainability and social good (Atif, 2023; Leng et al., 2022; Xu et al., 2021).

At the same time, critics warn of 'sustainability-washing': adopting 5.0 labels without material changes to production, procurement, or labor practices. Rebound effects (efficiency gains increasing total resource use), offshoring of carbon-intensive processes, and supply-chain externalities can blunt purported gains. Measuring 'well-being' remains contested, and ESG regimes are uneven in rigor, creating room for symbolic compliance. These concerns suggest the need for auditable metrics, independent verification, and just-transition provisions that make social and ecological commitments enforceable rather than aspirational.

For Education 5.0, sustainability and well-being enter the picture in multiple ways. Firstly, Education 5.0 aims to equip learners with the knowledge and values to address global sustainability challenges. This is often framed as *education for sustainable development* and global citizenship education, empowering students to contribute positively to society and the environment (Shahidi Hamedani et al., 2024; UNESCO, 2021b). Curricula are being updated in some places to include content on climate science, renewable energy, ethics of technology, and social entrepreneurship, reflecting the overlap of Education 5.0 with global priorities (Seeling et al., 2022). Secondly, the concept of well-being has been extended to education systems: Education 5.0 emphasizes inclusive and equitable learning environments, student well-being (mental health, emotional intelligence), and the idea that education's purpose is not just to produce workers but well-rounded individuals who can lead fulfilling lives and help their communities. This resonates with UNESCO's vision of a new social contract for education grounded in human dignity and sustainability. The UNESCO (2021) report *"Reimagining Our Futures Together"* argues that education must unite us collectively to forge sustainable futures for all. Initiatives under Education 5.0 include promoting green campuses, encouraging students to engage in community service and environmental projects, and ensuring digital transformation in education does not leave disadvantaged groups behind (addressing the digital divide).

Turning to Work 5.0, the inclusion of social well-being is evident in the focus on job quality and *"decent work."* Work 5.0 strategies advocate using technology to improve working conditions and make jobs more rewarding rather than simply more efficient (Anthuvan and Maheshwari, 2025; Rabhi et al., 2025). There is an emphasis on ergonomics, work-life balance, diversity and inclusion, and fair compensation. For example, some Work 5.0 discussions involve rethinking work hours (e.g., trials of four-day work weeks) to enhance the quality of life since automation could allow the same output with less human labor (ILO, 2022b). There is also attention on how to support workers through transitions, for instance, providing strong social safety nets and re-employment services for those displaced by automation so that the benefits of technological change are broadly shared (Chigbu and Nekhwevha, 2021, 2022; ILO, 2022a). The "human-centric approach" at the core of Work 5.0 means that productivity improvements are not pursued at the expense of worker rights or society's welfare. On the contrary, technology adoption is seen as an opportunity to design more *inclusive* workplaces, perhaps creating new roles that can be suitable for people with disabilities (with assistive tech) or enabling remote/flexible work that allows people to participate in the labor force under better conditions (something accelerated by the COVID-19 experience) (WEF, 2023, 2025).

In all, the shared focus on sustainability and well-being creates a strong value alignment across the three domains. It means, for instance, that Industry 5.0 initiatives naturally demand new educational content (Hashim et al., 2024) (so that future engineers and managers are trained in sustainable practices and ethics). They create new kinds of jobs (e.g., "sustainability specialists" or green tech jobs) for which education systems must prepare students (WEF, 2023, 2025). It also means that Work 5.0 policies (like emphasizing green jobs or care economy jobs) feed back into industry and education by shaping which sectors and skills are prioritized. This alignment is evident in international policy dialogues: the WEF's agenda, for example, combines discussions on the Future of Jobs with those on Stakeholder Capitalism and ESG (Environmental, Social, Governance) goals, reflecting that the future of work is inseparable from the goal of a sustainable, inclusive economy (WEF, 2025). As another example, the Bridges 5.0 project in the EU explicitly seeks to integrate Industry 5.0 and skills development with a focus on sustainability, indicating that policy initiatives are taking an integrated approach. See Fig. 4 for a *Venn diagram* that explicitly depicts domain-exclusive items, pairwise overlaps, and the three-way overlap across Industry 5.0, Education 5.0, and Work 5.0 (illustrative, not exhaustive).

We triangulate findings across empirical and technical studies on human-centric implementation. HDTs improve human-system integration and ergonomics, but standardization, data governance, privacy, and behavioral modeling remain open challenges (Bucci et al., 2025; Gaffinet et al., 2025; Wang et al., 2024). On shop-floor collaboration, architectures such as Industry 5.0 collaboration architecture (I5arc) clarify 'human-in-the-loop' control and role allocation yet reveal orchestration hurdles (work design, interfaces, oversight) that can negate expected productivity or satisfaction gains (Tóth et al., 2023). Empirical and evaluation evidence is not uniform: a multi-case multi-criteria decision-making (MCDM) study on Dynamic Capabilities 5.0 prioritizes collaborative ecosystems and digital readiness above people-centric innovation, challenging the assumption that 'human-centric' initiatives alone deliver resilience (Mohammed et al., 2025).

Implementing 5.0 faces technical, organizational, and economic challenges: interoperability and standards for HDTs, data and edge integration, and concerns around safety, privacy, and ethics (Bucci et al., 2025; Gaffinet et al., 2025; Wang et al., 2024); issues of role design, trust, governance, and change management (Fani et al., 2024; Tóth et al., 2023); and cost, SME constraints, skills gaps, and infrastructure limitations (Mohammed et al., 2025; UNESCO, 2021a; Zishiri et al.,

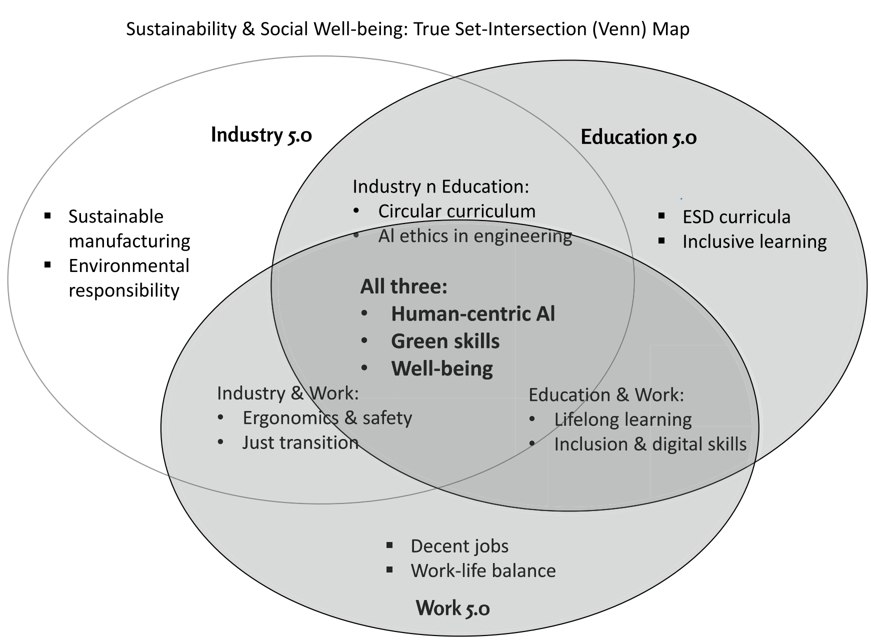

Fig. 4. Three-way overlap across Industry 5.0, Education 5.0, and Work 5.0 (illustrative, not exhaustive). Source: Authors.

2024). These barriers are especially pronounced in the Global South, where weak connectivity, limited training, and budget constraints impede Education 5.0 adoption (Zishiri et al., 2024), while UNESCO (2021a) notes that even in well-resourced systems, equity, privacy, and teacher-capacity gaps persist. Contradictions also surface across sectors: Lean-based transitions in fashion yield benefits but depend heavily on capability building and sustained change management (Fani et al., 2024).

Items are placed according to a set of logic. Domain-exclusive examples (e.g., *sustainable manufacturing; ESD curricula; decent jobs*) sit outside overlaps; pairwise intersections show joint priorities (e.g., *Industry and Education: circular curriculum, AI ethics in engineering; Industry and Work: ergonomics and safety, just transition; Education and Work: lifelong learning, inclusion and digital skills*); the triple intersection (*All three*) lists human-centric AI, green skills, and well-being as goals shared by Industry 5.0, Education 5.0, and Work 5.0. This figure is illustrative and complements the evidence discussed in the text. To summarize, sustainability and social well-being are not peripheral concerns but central pillars of Industry 5.0, Education 5.0, and Work 5.0. They provide a common language and goals that bind the domains together. By orienting technological and educational advancement toward these higher purposes, the "5.0" paradigm represents an attempt to ensure that the next wave of innovation contributes to solving humanity's most significant challenges (from climate change to inequality) rather than exacerbating them. In short, 5.0's normative shift is meaningful *only if* institutional checks convert principles into practice; absent, the language of well-being and sustainability risks legitimizing business-as-usual.

5. Discussion

5.1. Toward a unified 5.0 paradigm

The above thematic synthesis demonstrates that Industry 5.0, Education 5.0, and Work 5.0 are deeply interconnected facets of a broader transformation. By examining their intersections, we discern an emerging interdisciplinary 5.0 paradigm with significant implications for theory and practice. In this discussion, we reflect on these implications, consider how an integrated 5.0 approach can inform policy and organizational strategy, and identify challenges and tensions that must be navigated. The originality of this study lies in going beyond descriptive mapping to advance a novel conceptual synthesis: we explicitly position Industry, Education, and Work 5.0 as *mutually constitutive domains* whose interactions generate new theoretical insights (e.g., skills development as a cross-domain mediator, sustainability as a shared normative anchor, and AI ethics as a critical point of convergence). This framing contributes to scholarship by providing a unifying lens rather than treating each "5.0" strand in isolation. To avoid doubt, we reject a hierarchical "Education → Work → Industry" pipeline: our framework models the three as interdependent, overlapping subsystems connected by bidirectional flows of knowledge, skills, technology, and governance. We therefore treat skills development as an Education 5.0-anchored mediator enabling Work 5.0 transitions and Industry 5.0 capability, rather than a separate conceptual domain. As highlighted above, while synergies exist across Industry, Education, and Work 5.0, important contradictions remain: automation may still displace or deskill workers despite human-centric intentions, AI in education risks widening inequalities without strong governance and public investment, and sustainability goals can conflict with productivity imperatives through rebound effects or symbolic compliance.

5.2. Policy integration across industry, education, and labor

One key implication is the need for cross-sectoral policy alignment. Historically, industrial policy, educational policy, and labor policy have often operated in silos. Our review suggests that in the 5.0 era, such separation is increasingly untenable. For instance, a government aiming to foster human-centric innovation in industry (Industry 5.0) will likely fail if the education system is not simultaneously preparing a pipeline of appropriately skilled and innovative thinkers (Education 5.0) or if labor regulations do not support retraining and mobility for workers to move into new roles (Work 5.0). Policy bodies are starting to recognize this: the WEF implores stakeholders to invest in people through education and reskilling as the surest path to navigate the future of jobs (WEF, 2023, 2025), while the European Commission's Industry 5.0 agenda explicitly calls on industry to become a *"solution-provider for people and planet,"* which by definition involves contributions to societal goals beyond its immediate remit (European Commission, 2021). UNESCO's Futures of Education report likewise urges that we break down silos and jointly shape a better future through integrated efforts in education, science, and innovation (UNESCO, 2021b). These calls point toward a new paradigm of policy integration – a recognition that economic development, educational advancement, and labor welfare must be pursued in unison. This might mean forming inter-ministerial task forces on "Society 5.0" or "Human-Centric Digital Transformation" that coordinate initiatives across education, industry, and labor ministries. It could also mean aligning funding mechanisms: investing in AI R&D for industry, investing in AI training programs for workers, and updating school curricula on AI ethics.

5.3. Organizational leadership and strategic alignment

Another implication relates to organizational strategy and leadership in both industry and education. Embracing the 5.0 principles likely requires a cultural shift within organizations. In industry, leaders would need to balance technological adoption with human-centric design – possibly requiring new roles like *Chief Human-Centricity Officer* or *Ethics Officer* to ensure that worker well-being and sustainability are considered at the same level as financial metrics. Companies may also need to adopt more participatory approaches, involving employees in technology deployment decisions (as discussed under Work 5.0). In educational institutions, leadership would need to foster innovation in pedagogy and curriculum, encouraging educators to experiment with AI tools and interdisciplinary modules while safeguarding the core values of education. Notably, the concept of Education 5.0 challenges traditional academic departmental silos, pushing universities and schools to collaborate more with external partners (industry, communities) and to integrate fields (for example, combining engineering with ethics or computer science with social sciences). Engineering programs are already redesigning curricula to embed human-centric and sustainability principles for Industry 5.0 (Caratozzolo et al., 2024). This calls for visionary leadership that can navigate institutional inertia and drive change. Implications also extend to open innovation management: embedding open innovation principles (bounded rationality, social open innovation, and innovation limits) into Industry-Education-Work 5.0 strategies can guide policy and managerial practice (Bogers et al., 2018; Yun et al., 2020). For example, SME participation in open innovation consortia or platform-based ecosystems can diffuse human-centric practices and reduce the risk of innovation lock-in.

5.4. Ethical challenges in human-AI collaboration

Our review also highlights several challenges and tensions inherent in pursuing an integrated 5.0 agenda. One such tension is the ethical dimension of AI and data. If Industry 5.0 and Education 5.0 are to rely heavily on AI and data-driven personalization, questions of privacy, bias, and autonomy arise. For example, using AI tutors in classrooms can improve learning, but data about student performance must be handled ethically and transparently. Similarly, AI decision-support for workers can boost productivity, but if not implemented with care, it could introduce surveillance concerns or diminish worker autonomy. This suggests that as these paradigms advance, ethical frameworks and regulations must evolve in tandem. Interdisciplinary collaboration with ethicists, legal scholars, and social scientists will be important in formulating guidelines (e.g., on human-centric AI design, data governance, and algorithmic transparency) that uphold human rights and dignity. Bodies like UNESCO and the IEEE have started work on AI ethics in education and industry, providing principles that could guide practitioners (See: IEEE, 2019; UNESCO, 2022). Beyond privacy and bias, *allocative harms* (who benefits, who bears risk) and *epistemic opacity* (black-box models) complicate accountability. We therefore argue for minimum conditions for deployment: impact assessments before scale-up, contestability/appeals for algorithmic decisions, human-in-the-loop by design (with *meaningful* override), and negotiated data governance (purpose limitation, proportionality, retention).

5.5. Equity, inclusion, and the risk of uneven benefit

Another challenge is ensuring equity and inclusion. There is a risk that the benefits of the 5.0 transformations could accrue unevenly. For instance, high-tech industries might flourish in wealthy regions while others lag, or advanced education technology might benefit primarily well-resourced schools and learners. Work 5.0 emphasizes "good work" in general and good work *for all*. This requires deliberate measures to avoid widening gaps. It could involve, for example, public investments to bring cutting-edge educational technologies to rural or underserved areas and strong social policies (like upskilling programs, social protection floors, and job transition support) to help workers in at-risk jobs find better opportunities in the new economy. An integrative approach can help here: if industry, education, and labor stakeholders join forces on regional development initiatives, it might be possible to create local ecosystems where education programs are tailored to emerging industries and new jobs are created in areas needing renewal (e.g., training former coal industry workers to participate in renewable energy projects, combining sustainability, education, and new work). Distributional analysis should be integral to 5.0 programs (ex-ante). Without progressive financing and targeted support, reforms may concentrate advantages in already privileged regions/firms/schools. We highlight the importance of social dialogue (unions, parent bodies, community groups) and place-based industrial and education policy to counter agglomeration effects.

5.6. Regional variability and contextual differences

Implementation of Industry 5.0, Education 5.0, and Work 5.0 varies markedly across regions owing to differences in industrial structure, state capacity, fiscal space, digital and energy infrastructure, labour-market institutions, demographics, and social dialogue traditions. We therefore avoid universal prescriptions and articulate context-sensitive trajectories.

- o *European Union:* High regulatory capacity (e.g., European Commission's Industry 5.0), strong social partners, and green-transition funds enable augmentation-oriented automation, skills compacts, and ESG-aligned investment; constraints include SME retrofit costs and cross-border standards fragmentation (European Commission, 2021; WEF, 2023).
- o *Japan (Society 5.0):* Ageing demographics and an emphasis on societal harmony orient Work 5.0 toward care robotics and community-centric services; Education 5.0 emphasises human-AI teaming and ethics to sustain productivity and well-being (UNESCO, 2021b).
- o *Sub-Saharan Africa:* Education 5.0 policies (e.g., Zimbabwe) emphasize entrepreneurship, local industrialization, and inclusion; binding constraints include connectivity, device access, and teacher professional development, necessitating phased AI integration, public financing, and open educational resources (UNESCO, 2021a).
- o *South/Southeast Asia:* Large skilling missions and export-manufacturing/services ecosystems support Industry 5.0 adoption; risks include uneven quality assurance in micro-credentials and rural–urban divides without targeted investment (UNESCO, 2021a, 2021b).
- o *Latin America and the Caribbean:* High informality and platform work shape Work 5.0 priorities (social protection, formalization, digital rights), while industrial heterogeneity requires place-based Industry 5.0 strategies and public–private training partnerships (ILO, 2022a; WEF, 2023).

Across all regions, minimum safeguards including participatory design, human-in-the-loop oversight, auditable data governance, portability of learning rights, and just-transition financing mediate outcomes and reduce the risk of exacerbating inequality (IFOW, 2021; UNESCO, 2021b; WEF, 2023).

5.7. Lessons from the COVID-19 pandemic

The COVID-19 pandemic experience of 2020–2022 provides an illustrative backdrop that accelerated some of the 5.0 trends and exposed areas needing improvement. Remote work and hybrid education became widespread out of necessity, effectively pushing elements of Work 5.0 and Education 5.0 into the mainstream overnight. This showed the potential of technology to enable flexibility (many workers and students appreciated the new modes) but also highlighted issues like digital inequality (those without good internet or devices were disadvantaged) and the importance of human connection (the value of in-person interaction became evident when it was absent). The pandemic arguably strengthened the case for a human-centric, resilient approach: companies realized that employee well-being is critical to performance in crisis times, educators realized that technology is a tool, not a panacea, and policymakers saw the need for proactive planning for workforce shifts (as entire industries like hospitality or travel were disrupted). These lessons reinforce the ideas at the heart of Industry 5.0, Education 5.0, and Work 5.0, lending real-world support to what might have earlier seemed abstract concepts.

Practical constraints also loom large. Organizations face change-management fatigue; compliance and retrofit costs can be prohibitive for SMEs and under-funded schools; regulatory capacity to oversee AI and ESG varies widely; and vendor lock-in can undermine public value. Geopolitical fragmentation of standards further complicates cross-border supply chains and credential recognition. These frictions temper expectations and underscore that 5.0 outcomes are *conditional* on governance, financing, and institutional learning.

Finally, while our integrative framework is preliminary, it makes an original contribution by reframing the "5.0" conversation from parallel discourses into a unified paradigm with shared mediators, contradictions, and normative commitments. Rather than offering only a descriptive synthesis, we foreground analytical insights, such as the tension between human-centric rhetoric and persistent automation logics, or the uneven global applicability of 5.0 ideals. For practitioners and decision-makers, this means that adopting a siloed approach to innovation, education, or work will likely miss the mark in the coming decades. The framework presented here not only provides a common language (human-centric innovation, sustainability, augmentation) but also highlights *points of friction* where further theory, policy, and empirical research must intervene. By clarifying these intersections and tensions, this study advances the theoretical grounding of the 5.0 paradigm and sets a clearer research agenda for scholars across disciplines.

To make the framework actionable, we consolidate evidence-based recommendations across domains: (i) Education 5.0 should pursue curriculum co-design with industry, embed AI ethics and sustainability across subjects, and expand portable learning rights and micro-credentials (UNESCO, 2021a, 2021b; WEF, 2023); (ii) Work 5.0 requires negotiated worker-participation mechanisms, codetermination in AI deployment, and employer-led reskilling supported by public training subsidies (ILO, 2022a; OECD, 2023); and (iii) Industry 5.0 calls for participatory technology design, adoption of human-centric metrics (safety, well-being, inclusion), and auditable sustainability indicators with just-transition provisions (Fani et al., 2024; Mohammed et al., 2025). Together, these recommendations provide concrete guidance for policymakers, educators, and managers seeking to operationalize the 5.0 paradigm.

5.8. Limitations

This integrative review offers a broad synthesis of the emergent 5.0 paradigms but is subject to several limitations. First, as a qualitative synthesis, it may reflect selection and interpretation biases despite efforts to include diverse, reputable sources. Second, given the evolving nature of Industry 5.0, Education 5.0, and Work 5.0, some recent developments may not be fully represented, especially in generative AI and policy shifts. Third, the review prioritizes breadth over depth, covering multiple disciplines at the expense of domain-specific granularity. Fourth, the reliance on predominantly English-language and high-profile publications may underrepresent grassroots and non-Western perspectives, such as localized applications of Society 5.0. Fifth, much of the literature in this space remains conceptual, with limited empirical validation; thus, findings should be interpreted as indicative rather than conclusive. Lastly, terminological differences across disciplines may lead to oversimplification or loss of context. These limitations underscore the need for future empirical studies, context-specific analyses, and expanded global engagement to refine the frameworks presented here. Our country-level depth is limited; future work should expand comparative, context-specific studies.

5.9. Future research directions

Given the nascent stage of the integrated “5.0” discourse, there are ample opportunities for future research to expand knowledge and inform practice. We outline several promising directions below.

5.9.1. Empirical validation of 5.0 outcomes

Much of the current literature on Industry 5.0, Education 5.0, and Work 5.0 is conceptual. Empirical studies are needed to validate whether implementing these paradigms delivers the promised benefits (and under what conditions). For example, researchers could conduct pilot intervention studies in education, such as selecting a group of schools implementing Education 5.0 methods (project-based learning, AI tutors, interdisciplinary sustainability projects, etc.) and comparing outcomes (student engagement, skill acquisition, creativity, academic performance) with traditional schools over several years. Similarly, in industry, comparative case studies of companies that embrace Industry 5.0 practices (human-centric cobots, green manufacturing processes, participatory workplace design) versus those that stick to a more traditional Industry 4.0 approach could reveal differences in productivity, innovation rates, worker retention, and well-being, etc. On the workforce side, experiments like trials of reduced work weeks or “right to disconnect” policies in the spirit of Work 5.0 could be evaluated for their impacts on employee well-being and company performance. Such empirical evidence would help build the business or educational case for the 5.0 transformations. It would also identify any unintended consequences early, allowing strategies to be refined.

5.9.2. Longitudinal and foresight studies

The transformations envisaged by Industry 5.0, Education 5.0, and Work 5.0 are long-term and dynamic. Longitudinal research can track changes over time to see how trajectories unfold. For instance, a longitudinal study could follow a cohort of students who experience an Education 5.0-aligned curriculum and see how they fare in the labor market 5–10 years after graduation: Are they more adaptable to job changes? Do they attain “good work” outcomes? At a macro level, foresight and scenario planning studies could be valuable. Researchers might construct scenarios for, say, the year 2035 or 2040 under different levels of adoption of 5.0 principles (e.g., one scenario where human-centric, sustainable approaches are widely adopted across sectors and another where the world remains stuck in 4.0-style automation-first logic). By comparing these scenarios, we can better anticipate potential benefits, risks, and requirements. Such studies often use mixed methods (expert panels, simulations, trend analysis). They can guide policymakers by highlighting, for example, that in a high-adoption scenario, we might see higher societal well-being but new governance challenges (like data governance needs). In contrast, we might see higher inequality or public backlash against technology in a low-adoption scenario. This kind of future research ensures that we remain prepared for various outcomes as we push forward.

5.9.3. Policy implementation research

As governments and international organizations start to incorporate 5.0 ideas into strategies (e.g., the European Commission integrating Industry 5.0 into its industrial strategy or UNESCO promoting Education 5.0 concepts in global forums), it is important to study how these policies are implemented and with what effect. Comparative policy analysis could examine different national approaches: for instance, how Japan’s “Society 5.0” policy (which overlaps significantly with these concepts) is being executed, and what can be learned from it. How does it compare to the EU’s approach or smaller countries like Singapore’s educational reforms? Research can also identify barriers to implementation, be they bureaucratic, cultural, financial, or political. Understanding these can help refine strategies. For example, suppose a study finds that a lack of teacher training is hampering the Education 5.0 rollout in a specific context. In that case, that points to a need for investment in professional development. Suppose the industry finds that adopting human-centric practices costs more upfront and thus faces resistance. That might suggest a role for incentives or subsidies. Essentially, implementation science for the 5.0 agenda would help move ideas from paper to practice effectively.

5.9.4. Cross-cultural and regional studies

The 5.0 paradigms have emerged mainly from specific contexts (Industry 5.0 was heavily driven by the EU; Education 5.0 ideas appear in places like Asia and Africa under different interpretations; Work 5.0 has roots in UK policy circles via IFOW). Future research should explore how these concepts translate to different cultural and regional settings. It is possible, even likely, that “human-centric innovation” will have different emphases in different cultures. For example, Japan’s Society 5.0 strongly emphasises using technology to assist an aging population and the concept of societal harmony. This might give their version of Work 5.0 a particular flavor (more focus on care robots or community cohesion). In developing countries, Education 5.0 might emphasize digital leapfrogging and inclusion (as seen in Zimbabwe’s focus on education for local industrialization [Andres et al., 2022]). By conducting case studies or comparative analyses in diverse settings, such as a Scandinavian country, a Sub-Saharan African country, and an East Asian country, researchers can enrich the global narrative. This will ensure that the integrated 5.0 framework is not one-size-fits-all but is informed by plural perspectives and that best practices are shared globally.

5.9.5. Technological developments and ethics

The pace of technological change means innovations will continually test the principles of the 5.0 paradigm. Research at the intersection of technology and human-centric design needs to keep up. For instance, as AI becomes more powerful (e.g., generative AI, autonomous systems), what does Industry 5.0 look like when AI can potentially take on creative tasks? How do we ensure human oversight and meaningful roles for people? In education, if brain-computer interfaces or neurotechnology become viable for learning enhancement, how do those align or conflict with the human-centric ethos of Education 5.0? These emerging tech areas (the *metaverse*, biotechnology, advanced robotics, etc.) should be examined through the lens of 5.0 values. Interdisciplinary research teams, including technologists, ethicists, and social scientists, could proactively develop "responsible innovation" frameworks that extend the human-centric, sustainable approach into these new domains. For example, one could articulate design guidelines for virtual reality in classrooms to keep it pedagogy-first and socially inclusive or for AI in hiring such that it supports fairness and diversity (an aspect of Work 5.0). The ethical dimension needs attention, ensuring privacy, preventing bias, and protecting human agency so that the 5.0 vision does not get undermined by legitimate public concerns about technology misuse.

5.9.6. Micro-level organizational and behavioral research

While our review has often taken a macro perspective, there is also a need for micro-level studies on implementing and sustaining 5.0 principles within organizations (be it a factory, a school, or a company). For the industry, research could focus on change management: What are effective strategies for a manufacturing firm to transition from a traditional model to an Industry 5.0 model? How can leadership overcome employee skepticism or retrain workers for new collaborative roles? What key performance indicator adjustments or incentive structures help embed human-centric values? Similar micro-level questions arise in education: How do teachers perceive AI tools, friend or foe, and what professional development helps them embrace Education 5.0 pedagogies? What classroom management techniques work when using immersive tech, etc.? How do teams build trust in AI systems in workplaces, and how can interface design or management interventions enhance human-AI teamwork? These granular questions are important, as they often determine whether grand ideas succeed or fail.

5.9.7. Evaluation metrics and analytics

Lastly, future research should also consider measuring success in the 5.0 context. Traditional metrics may not fully capture what we care about (e.g., GDP does not measure well-being; test scores do not equal creativity). There is room for developing new indices or assessment tools: perhaps an "Industry 5.0 index" that evaluates industries on human-centric and sustainability criteria or new metrics for schools that gauge how well they foster collaboration, ethical thinking, and academic knowledge. Big data and analytics might be leveraged to monitor progress on key outcomes (for instance, analyzing job posting data to see if demand for specific human-centric skills is rising or using sensors in smart factories to measure improvements in ergonomics and safety). Researchers can contribute by proposing and validating such multidimensional metrics, which can guide practitioners and policymakers in tracking the impact of 5.0 initiatives.

Overall, future research should be as integrative and forward-looking as the subject matter itself. A combination of empirical validation, strategic foresight, policy analysis, cross-cultural exploration, tech-and-ethics inquiry, micro-level organizational study, and metric development will all be valuable. The fifth revolution we are discussing is not a single-field issue; it lives at the intersection of engineering, management, education, sociology, public policy, and beyond. Therefore, researchers must collaborate across disciplines to fully understand and guide it. By doing so, the academic community can play a crucial role in shaping an inclusive and evidence-based path for Industry 5.0, Education 5.0, and Work 5.0 as they move from vision to reality.

6. Conclusion

This integrative review demonstrates that Industry 5.0, Education 5.0, and Work 5.0 converge as a human-centric, AI-enabled, and sustainability-driven paradigm. Analysis of 62 high-quality studies reveals three central findings: (1) human–AI collaboration reframes automation as augmentation rather than substitution, (2) skills development pathways led by Education 5.0 connect curricula, labor mobility, and lifelong learning, and (3) sustainability transitions act as a unifying anchor across economic, social, and ecological dimensions. At the same time, contradictions persist, including risks of job displacement, digital inequality, and rebound effects, as productivity goals conflict with sustainability outcomes.

The primary implication is that governments, industries, and educational institutions must align policies, curricula, and labor protections to enable just and inclusive 5.0 transitions. For organizations, this means embedding human-centric performance indicators, ethical AI governance, and sustainability metrics into strategy. For education providers, it calls for integrated, future-oriented curricula and open innovation partnerships that equip learners with skills for both digital and green economies.

Future research should move beyond conceptual synthesis by testing this framework through comparative case studies, developing standardized metrics for human-centric value creation, and analyzing how open innovation ecosystems (e.g., SMEs, platforms, collective intelligence) can accelerate Industry, Education, and Work 5.0 implementation globally.

CRediT authorship contribution statement

Sicelo Leonard Makapela: Writing – review & editing, Writing – original draft, Validation, Supervision, Resources, Conceptualization. **Bianca Ifeoma Chigbu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Ethical statement

The authors affirm that this manuscript, titled *"AI in Education, Sustainability, and the Future of Work: An Integrative Review of Industry 5.0, Education 5.0, and Work 5.0"*, is an original work and has not been published elsewhere, nor is it under consideration by any other journal. All sources of information and data used in the review have been appropriately cited to ensure academic integrity and transparency.

As this is an integrative literature review, it does not involve human participants, animals, or primary data collection; therefore, ethical approval and informed consent were not required. The review was conducted in accordance with the ethical guidelines for scholarly research and publication.

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The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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